

DSPO: STABLE AND EFFICIENT POLICY OPTIMIZATION FOR AGENTIC SEARCH AND REASONING

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ABSTRACT

Endowing large language models (LLMs) with the capacity to actively retrieve external knowledge is essential for the complex, real-world tasks that exceed the reach of static parametric memory. Existing approaches confront a persistent dilemma: prompting-based methods elicit a model’s latent agentic behavior without improving it, whereas reinforcement learning (RL), once transplanted to open-ended interactive tasks, is beset by performance ceilings and catastrophic training collapse that leave this agentic potential largely untapped. We address this dilemma with **Dynamic-filter Sequence-level Policy Optimization (DSPO)**, an RL algorithm engineered for robust agent training that couples sequence-level policy optimization with dynamic outcome-based sample filtering. DSPO trains the policy entirely through RL to interleave multi-turn search and reasoning, dispensing with supervised demonstration data altogether. Under a controlled protocol in which every method shares the same training data, BM25 retriever, and reward, our 7B model improves over a comparably sized Search-R1 baseline by **34.1%** in average subEM across seven question-answering benchmarks and, on demanding multi-hop QA such as HotpotQA, surpasses the 14B model of that prior work by nearly **9% relative** (0.613 vs. 0.563)—all while sustaining stable training precisely where token-level baselines collapse.

1 INTRODUCTION

Large Language Models (LLMs) (Brown et al., 2020; Touvron et al., 2023; Zhao et al., 2023) have attained exceptional performance across a broad spectrum of specialized tasks, spanning mathematics (Shao et al., 2024; Trinh et al., 2024; Yu et al., 2025), coding (Zheng et al., 2023; Yang et al., 2024), and creative writing (Chakrabarty et al., 2024; Marco et al., 2024). A fundamental limitation nonetheless persists: their knowledge is inherently static, bounded by the corpus on which they were trained. The most compelling remedy to this knowledge cutoff is to endow LLMs with search capabilities, transforming them into agents that actively query external knowledge sources (Jin et al., 2025b). Such retrieval-augmented interaction is a canonical instance of tool use (Schick et al., 2023), in which the model learns to invoke an external retrieval tool to resolve problems that lie beyond its parametric memory. Mastering this competence demands a multi-step policy that decides both what to reason about and when to query the environment, which casts the task as a sequential decision-making problem—precisely the regime for which Reinforcement Learning (RL) is best suited.

Unlike Supervised Fine-Tuning (SFT), which depends on costly static demonstrations and cannot teach exploration, RL furnishes a principled framework in which LLMs acquire effective policies through trial-and-error interaction (Chu et al., 2025). Value-free methods such as Group Relative Policy Optimization (GRPO) (Shao et al., 2024) have accordingly become the dominant paradigm, prized for their simplicity and reduced memory footprint. Yet a method that succeeds on more constrained, single-turn tasks does not automatically transfer to the open-ended domain of interactive search, where GRPO exhibits critical instabilities (Jin et al., 2025b; Yu et al., 2025; Cui et al., 2025; Liu et al., 2025). In our experiments this fragility traces to two mutually compounding failure modes. First, as identified by Zheng et al. (2025), GRPO’s token-level objective is mismatched to a sequence-level reward, producing high-variance gradient estimates that accumulate over long trajectories and destabilize training. Second, the sparse binary rewards intrinsic to search tasks frequently yield rollout groups with homogeneous outcomes—uniformly successful or uniformly unsuccessful—whose normalized advantage collapses to zero, depriving the optimizer of any usable gradient and severely curtailing sample efficiency (Yu et al., 2025; Liu et al., 2025). Because these two pathologies reinforce one another over the long horizons that agentic reasoning entails, addressing either alone leaves the other free to derail training.

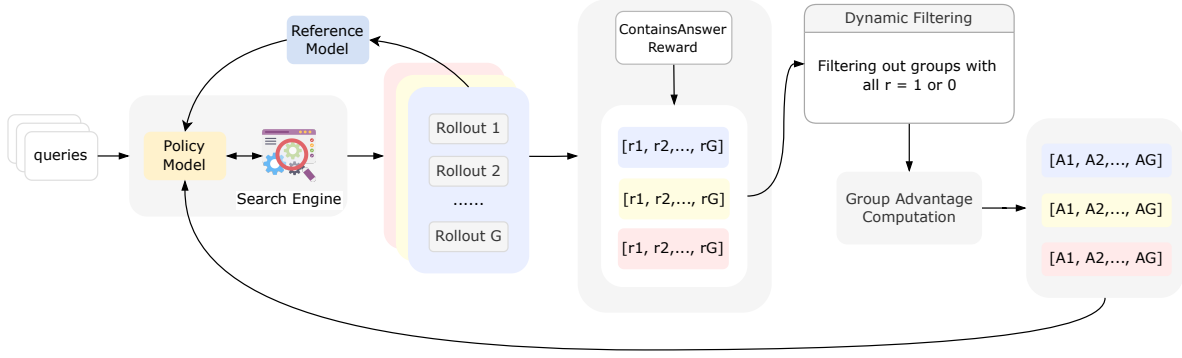


Figure 1: Overview of the DSPO training loop. For a given query, the policy model generates a group of G trajectories by interleaving reasoning with calls to the search environment, and each trajectory receives a sparse terminal reward. The **dynamic filter** discards groups with homogeneous outcomes—those that would yield a degenerate, zero-variance advantage—and continues sampling until the batch is filled, so that every training batch supplies an informative advantage signal. Group-relative advantages are then computed and used to update the policy through the sequence-level objective.

To confront these two failure modes jointly, we introduce **Dynamic-filter Sequence-level Policy Optimization (DSPO)**, an algorithm that integrates two complementary principles from recent policy optimization research into a single framework tailored to the demands of agentic search. To resolve the objective–reward mismatch, DSPO performs optimization at the sequence level in the spirit of GSPO (Zheng et al., 2025), matching the unit of optimization to the unit of reward and thereby stabilizing learning over long horizons. To resolve the vanishing-signal problem, DSPO couples this objective with a dynamic outcome-based filtering mechanism inspired by DAPO (Yu et al., 2025) that constructs each training batch exclusively from rollout groups exhibiting both successful and unsuccessful outcomes for a given prompt, guaranteeing that the advantage signal $\hat{A}_i$ remains well-defined and informative. Neither ingredient is individually novel; our contribution is to show that they are separately necessary and jointly sufficient for stable, efficient training in the agentic search setting, and to characterize why—a finding that our ablations make precise. Under a controlled protocol that holds the training data, retriever, and reward fixed across methods, the resulting 7B model attains a **34.1% relative improvement** in average subEM over an equally sized Search-R1 baseline (Jin et al., 2025b) and, on demanding multi-hop benchmarks such as HotpotQA, surpasses that work’s 14B counterpart (Jin et al., 2025a) by **nearly 9% relative** (0.613 vs. 0.563).

Our main contributions are as follows:

- **A unified algorithm for stable agentic RL.** We propose DSPO, which addresses the coupled instability and sample-inefficiency that impede the training of agentic search models by combining two mutually reinforcing principles within a single framework: **sequence-level optimization**, which yields robust policy updates by matching the optimization unit to the sequence-level reward, and **dynamic outcome-based filtering**, which sustains a dense, informative learning signal under sparse rewards. Our claim is not that either principle is new, but that their integration is what makes stable, efficient training of multi-turn search agents achievable.
- **Strong performance under a controlled, like-for-like evaluation.** Across seven QA benchmarks, and with every compared method sharing the same training data, BM25 retriever, and reward, our 7B model achieves a **34.1% relative improvement** in average subEM over an equally sized Search-R1 baseline and, on complex multi-hop QA, matches or exceeds its 14B counterpart—attaining a nearly **9% relative gain** on HotpotQA (0.613 vs. 0.563). That a 7B policy overtakes a 14B one under matched conditions indicates that the gains stem from the optimization algorithm rather than from model scale.
- **Stability isolated to the optimization algorithm.** Through component ablations and training-curve analysis, we show that DSPO follows a smooth, near-monotonic learning trajectory precisely where token-level baselines collapse. Because all runs use a single basic BM25 retriever, the improvements are attributable to the robustness of the optimization algorithm rather than to retrieval quality, and we further find that DSPO’s principles transfer to single-turn mathematical reasoning.

2 RELATED WORK

2.1 RL FOR LLMs

The landscape of RL for LLMs has evolved rapidly, progressing from foundational Reinforcement Learning from Human Feedback (RLHF) methods built on PPO and explicit reward models (Ouyang et al., 2022; Christiano et al., 2017; Schulman et al., 2017) toward simpler direct-optimization frameworks such as DPO (Rafailov et al., 2023). A decisive shift toward value-free optimization arrived with Group Relative Policy Optimization (GRPO) (Shao et al., 2024), which streamlines training by deriving the advantage signal directly from group statistics rather than from a learned critic. GRPO’s token-level objective, however, is a documented source of training instability (Liu et al., 2025; Cui et al., 2025), prompting a series of targeted refinements: GSPO recasts the objective at the sequence level to match the unit of reward (Zheng et al., 2025), DAPO counters inefficient learning under sparse rewards through a dynamic outcome-based sampling mechanism (Yu et al., 2025), and GMPO stabilizes the token-level objective via a geometric-mean aggregation that dampens sensitivity to outliers (Zhao et al., 2025). These refinements each target a single pathology, and our experiments indicate that, applied in isolation to the open-ended search setting, each still succumbs to training collapse or plateaus at a suboptimal ceiling. DSPO builds directly on this line of work: rather than proposing a new optimization primitive, it composes sequence-level optimization with dynamic outcome-based filtering into one algorithm and demonstrates that this specific pairing is what the distinctive challenges of training autonomous search agents require.

2.2 LLMs WITH AGENTIC RETRIEVAL

To alleviate the static-knowledge limitation of LLMs, retrieval-augmented generation (RAG) couples the model with external retrievers that dynamically inject evolving information (Lewis et al., 2020; Gao et al., 2023). Classical RAG pipelines employ dense retrievers to fetch relevant documents, which are then concatenated into the LLM’s input for generation (Karpukhin et al., 2020); because the retrieval step is fixed and executed once, however, such pipelines afford little autonomy in complex, multi-turn scenarios. Research has consequently shifted toward agentic paradigms in which the LLM acts as an autonomous agent that plans, searches, and reasons iteratively. ReAct interleaves reasoning and acting so that LLMs can operate tools for tasks such as web navigation (Yao et al., 2023), while multi-agent systems such as AutoGen orchestrate collaborative workflows (Wu et al., 2024). A further line of work fuses agentic RAG with RL so that agents refine their retrieval behavior through learned decision-making: Wu et al. (2025) introduce Agentic Reasoning, which integrates external tools to streamline LLM reasoning, and several RL-based approaches (Jin et al., 2025b; Chen et al., 2025; Song et al., 2025) train LLMs to interleave reasoning and search end-to-end from reward alone. This end-to-end paradigm internalizes agentic competence within a single model and sidesteps the engineering overhead of multi-agent frameworks, yet it remains hampered by the training instability and performance ceilings characteristic of the open-ended search domain. DSPO targets exactly this bottleneck, supplying a training procedure whose stability enables LLMs to acquire effective multi-turn search strategies without supervised demonstrations.

3 METHODOLOGY

This section develops DSPO in three steps. We first formulate agentic search as an RL problem and review the prior policy optimization algorithms on which we build, making explicit the two limitations that render them inadequate in this setting. We then introduce **Dynamic-filter Sequence-level Policy Optimization (DSPO)**, specifying the two components through which it secures training stability and sample efficiency and the precise failure mode each addresses. Finally, we present the integrated objective and training procedure that unify them.

3.1 PRELIMINARIES

Policy Gradient Methods for LLMs. Training LLMs via RL commonly relies on policy gradient methods such as PPO (Schulman et al., 2017), a workhorse for LLM alignment. PPO optimizes a policy π_θ by maximizing a clipped surrogate objective computed from samples drawn under a behavior policy $\pi_{\theta_{\text{old}}}$. Averaged over tokens, the objective is:

$$J_{\text{PPO}}(\theta) = \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi_{\theta_{\text{old}}}(\cdot|x)} \left[\frac{1}{|y|} \sum_{t=1}^{|y|} \min \left(r_t(\theta) \hat{A}_t, \text{clip} \left(r_t(\theta), 1 - \epsilon, 1 + \epsilon \right) \hat{A}_t \right) \right], \quad (1)$$

where $r_t(\theta) = \frac{\pi_\theta(y_t|x, y_{<t})}{\pi_{\theta_{\text{old}}}(y_t|x, y_{<t})}$ denotes the token-level importance ratio. PPO, however, requires a separately trained value model to estimate token-level advantages $\hat{A}_t$ via Generalized Advantage Estimation (GAE) (Schulman et al., 2015); this critic incurs substantial memory overhead and can itself become a source of instability.

GRPO (Shao et al., 2024) removes this dependence. In place of a learned value model, GRPO samples a group of G responses $\{y_i\}_{i=1}^G$ for each prompt x and estimates each response’s advantage by normalizing its reward against the group statistics. As in PPO, optimization proceeds at the token level:

$$J_{\text{GRPO}}(\theta) = \mathbb{E}_{x \sim \mathcal{D}, \{y_i\}_{i=1}^G \sim \pi_{\theta_{\text{old}}}(\cdot|x)} \left[\frac{1}{G} \sum_{i=1}^G \frac{1}{|y_i|} \sum_{t=1}^{|y_i|} \min \left(r_{i,t}(\theta) \hat{A}_i, \text{clip}(r_{i,t}(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_i \right) - \beta D_{KL}(\pi_\theta || \pi_{\text{ref}}) \right], \quad (2)$$

where $r_{i,t}(\theta) = \frac{\pi_\theta(y_{i,t}|x, y_{i,<t})}{\pi_{\theta_{\text{old}}}(y_{i,t}|x, y_{i,<t})}$, and the advantage assigned to every token $y_{i,t}$ within a response y_i is the shared sequence-level quantity:

$$\hat{A}_{i,t} = \hat{A}_i = \frac{R_i - \text{mean}(R)}{\text{std}(R)}, \quad (3)$$

The essential observation is that every token of a response y_i is assigned the identical advantage $\hat{A}_i$, itself derived from a reward defined only at the sequence level, while the importance weight $r_{i,t}(\theta)$ that multiplies it varies token by token. This discrepancy—a sequence-level reward propagated through a token-level weighting—is precisely the mismatch that DSPO is designed to resolve.

Agentic Search as a Markov Decision Process. We formalize the iterative process of agentic search and reasoning as a discrete-time, finite-horizon Markov Decision Process (MDP), the components of which we specify as follows.

- **State (s_t):** At turn t , the state s_t encodes the complete interaction history: the initial question q , all previously generated thoughts and search queries, and the evidence retrieved by the environment. Because each retrieved result must be interpreted and integrated into subsequent decisions, the state accumulates monotonically over the trajectory, inducing the long-horizon dependencies that conventional token-level RL is ill-equipped to optimize.
- **Action (a_t):** An action is a complete textual segment produced by the policy π_θ , comprising free-form reasoning followed by a terminal decision. The segment terminates either with a `</tool_call>` token, which issues a search whose results are returned within `</tool_response>`, or with a `</answer>` token, which ends the trajectory. This structured action space requires the agent to learn not only *what* to generate but also *when* to search—a coupling that induces substantial variability in trajectory length and thereby motivates length-invariant optimization.
- **Policy (π_θ):** The policy π_θ is the underlying LLM, which generates tokens autoregressively conditioned on the state. Although the policy conditions on the full history, the optimization target is evaluated only over model-generated thoughts and actions, with environment-returned content masked out (Jin et al., 2025b). Optimizing π_θ therefore requires credit assignment across long sequences in which many intermediate reasoning steps receive no direct reward, which underscores the need for a stable sequence-level update.
- **Trajectory (τ):** A trajectory $\tau = (s_1, a_1, \dots, s_T, a_T)$ records the full sequence of reasoning and tool interactions for a given question, comprising both model-generated actions and environment-returned search results. Because the reward is assigned to the trajectory as a whole, the optimization problem is intrinsically sequence-level: any early decision may propagate to the eventual correctness of the answer.
- **Reward ($R(\tau)$):** We adopt a sparse terminal reward: a trajectory receives $R = 1$ when its final answer contains the ground-truth text and $R = 0$ otherwise,

$$R(\tau) = \begin{cases} 1 & \text{if } a_{\text{gold}} \subseteq a_{\text{pred}}, \\ 0 & \text{otherwise.} \end{cases} \quad (4)$$

RL with a Search Engine. Following Jin et al. (2025b), we model the search engine $\mathcal{S}$ as an explicit component of the environment. The policy LLM π_θ learns to generate trajectories by interleaving reasoning with calls to $\mathcal{S}$, and the

objective is to recover a policy that maximizes expected reward while a KL-divergence term regularizes against large deviations from a reference policy π_{ref} :

$$\max_{\pi_{\theta}} \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi_{\theta}(\cdot|x; \mathcal{S})} [R(x, y)] - \beta D_{\text{KL}} [\pi_{\theta}(y|x; \mathcal{S}) || \pi_{\text{ref}}(y|x; \mathcal{S})]. \quad (5)$$

Here, $y \sim \pi_{\theta}(\cdot|x; \mathcal{S})$ signifies that the trajectory y arises from a multi-step process that couples the policy’s token generation with the information returned by the search engine $\mathcal{S}$.

Motivation for DSPO. Although frameworks such as Search-R1 (Jin et al., 2025b) successfully cast agentic search as an RL problem, applying conventional algorithms such as PPO or GRPO in this setting encounters two compounding obstacles that the open-ended search environment aggravates. The first is the structural mismatch between the sequence level at which reward is assigned and the token level at which optimization is performed (Zheng et al., 2025); this discrepancy produces high-variance gradient estimates that accumulate over long trajectories and, in our runs, frequently culminate in policy collapse. The second is the sparsity of the binary reward: when a rollout group contains only successful or only unsuccessful trajectories, its normalized advantage degenerates and conveys no learning signal, which drastically reduces sample efficiency (Yu et al., 2025; Liu et al., 2025). Because the two failure modes interact—instability lengthens the search for informative gradients while sparse signal starves the very updates that could stabilize the policy—DSPO is designed to neutralize them jointly rather than in sequence.

3.2 DYNAMIC-FILTER SEQUENCE-LEVEL POLICY OPTIMIZATION

DSPO combines two coordinated mechanisms: (1) it performs policy optimization at the sequence level, aligning the unit of the training objective with the trajectory-level reward structure, and (2) it applies a dynamic filtering rule that admits for training only rollout groups carrying a non-degenerate learning signal. These components address, respectively, the objective–reward mismatch and the sparse-signal degeneracy identified above, and their integration constitutes the full training process depicted in Figure 1.

3.2.1 SEQUENCE-LEVEL POLICY OPTIMIZATION FOR ENHANCED STABILITY

Following GSPO (Zheng et al., 2025), we replace the token-level importance ratio with a length-normalized sequence-level counterpart. The sequence-level importance ratio $s_i(\theta)$ for a response y_i is defined as the geometric mean of its constituent token-level ratios, equivalently the exponential of their mean log-ratio:

$$s_i(\theta) = \left(\frac{\pi_{\theta}(y_i|x)}{\pi_{\theta_{\text{old}}}(y_i|x)} \right)^{\frac{1}{|y_i|}} = \exp \left(\frac{1}{|y_i|} \sum_{t=1}^{|y_i|} \log \frac{\pi_{\theta}(y_{i,t}|x, y_{i,<t})}{\pi_{\theta_{\text{old}}}(y_{i,t}|x, y_{i,<t})} \right). \quad (6)$$

The exponent $1/|y_i|$ is what distinguishes this ratio from the raw trajectory likelihood ratio $\pi_{\theta}(y_i|x)/\pi_{\theta_{\text{old}}}(y_i|x)$: taking the geometric mean rather than the product converts a quantity whose magnitude scales multiplicatively with sequence length into one that occupies a consistent numerical range independent of $|y_i|$. This length invariance is what makes a single clipping range meaningful across trajectories that vary widely in length, and it is the property on which the stability of the update rests. We do not claim that $s_i(\theta)$ equals the trajectory likelihood ratio; rather, it is a length-normalized surrogate whose empirical effect—illustrated by the gradient analysis below and the training curves in Section 4—is to curb the gradient variance that destabilizes token-level updates in this setting.

Gradient Analysis. Comparing the two gradients makes the source of DSPO’s stability explicit. The gradient of the unclipped token-level GRPO objective scales each token’s log-probability gradient by a noisy, token-specific weight $r_{i,t}(\theta)$, so per-token estimation noise enters the sum at every position. The gradient of our sequence-level objective, by contrast, scales the aggregate log-probability gradient of the whole sequence by a single, comparatively stable weight $s_i(\theta)$:

$$\nabla_{\theta} J_{\text{GRPO}} \propto \mathbb{E} \left[\hat{A}_i \cdot \sum_{t=1}^{|y_i|} r_{i,t}(\theta) \nabla_{\theta} \log \pi_{\theta}(y_{i,t} | \dots) \right] \quad (7)$$

$$\nabla_{\theta} J_{\text{DSPO}} \propto \mathbb{E} \left[\hat{A}_i \cdot s_i(\theta) \cdot \sum_{t=1}^{|y_i|} \nabla_{\theta} \log \pi_{\theta}(y_{i,t} | \dots) \right] \quad (8)$$

By applying one correction factor to the trajectory as a whole rather than a separate noisy weight at each position, the sequence-level update forestalls the accumulation of token-level variance that destabilizes prior methods, producing

markedly more stable training. This stability is a necessary but not sufficient condition for effective learning: as the next component makes clear, the dynamic filtering mechanism supplies the second condition by ensuring that the advantage signal $\hat{A}_i$ multiplying $s_i(\theta)$ is itself well-defined, assembling each batch from rollout groups that contain both successes and failures and thereby removing wasted samples in sparse-reward environments.

3.2.2 DYNAMIC OUTCOME-BASED FILTERING FOR EFFICIENT LEARNING

The sparse binary structure of our reward poses a direct challenge for group-based advantage estimation. When all G responses in a group are uniformly correct ($R = 1$) or uniformly incorrect ($R = 0$), the group has zero reward variance, so the normalized advantage $\hat{A}_i$ vanishes or is undefined; such degenerate groups contribute no gradient and squander the rollouts spent producing them.

DSPO addresses this failure mode with a dynamic filtering rule adapted from DAPO (Yu et al., 2025). During sampling we retain only those groups whose trajectories exhibit a mixture of successful and unsuccessful outcomes: a group $\{y_i\}_{i=1}^G$ is admitted for training if and only if its rewards $\{R_i\}_{i=1}^G$ satisfy

$$0 < \sum_{i=1}^G R_i < G. \quad (9)$$

The condition enforces strictly positive reward variance within every admitted group and thereby guarantees a well-defined, informative advantage. Sampling continues, discarding homogeneous-outcome groups, until the batch is filled; the procedure thus curates for each policy update a set of prompts that are neither trivially solved nor uniformly failed, converting an otherwise sparse reward into a consistently informative learning signal. This selectivity comes at the cost of additional rollouts spent on discarded groups, a trade-off we adopt because, as our ablations show, the resulting gains in signal density more than offset the extra sampling.

3.3 THE DSPO OBJECTIVE AND TRAINING ALGORITHM

Integrating the two components yields the final DSPO objective. For each admitted group drawn from the filtered sample space $\mathcal{D}_{\text{filtered}}$, we compute the advantage $\hat{A}_i$ by group-relative normalization:

$$\hat{A}_i = \frac{R_i - \text{mean}(R)}{\text{std}(R) + \delta}, \quad (10)$$

where δ is a small constant ensuring numerical stability. The policy π_θ is updated by maximizing:

$$J_{\text{DSPO}}(\theta) = \mathbb{E}_{(x, \{y_i\}) \in \mathcal{D}_{\text{filtered}}} \left[\frac{1}{G} \sum_{i=1}^G \min \left(s_i(\theta) \hat{A}_i, \text{clip}(s_i(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_i \right) \right], \quad (11)$$

where s_i is the sequence-level importance ratio of Eq. 6. During likelihood computation we mask the loss on every token returned by the search tool, following Jin et al. (2025b), so that the gradient acts only on the model’s own reasoning and query decisions and the policy is trained to exploit retrieved evidence rather than to reproduce it verbatim. The complete training procedure is specified in Algorithm 1.

4 EXPERIMENTS

We now present a systematic empirical evaluation of DSPO, organized to substantiate three claims: that, under a controlled protocol in which every method shares the same data, retriever, and reward, DSPO (1) attains the strongest performance across a suite of challenging question-answering benchmarks; (2) exhibits substantially enhanced training stability, avoiding the catastrophic collapse that afflicts token-level baselines; and (3) derives these gains from the synergy of its two components rather than from either alone. We additionally probe how the gains scale with model size and whether they transfer beyond agentic search.

4.1 EXPERIMENTAL SETUP

Following Search-R1 (Jin et al., 2025b), we adopt the prompt template shown in Table 1, which structures the model’s actions by delimiting reasoning, retrieval, and answer generation with the `<think>`, `<tool_call>`, and `<answer>` tags, respectively.

Algorithm 1 Dynamic-filter Sequence-level Policy Optimization (DSPO)

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1: Input: Initial policy  $\pi_{\theta_0}$ , fixed reference policy  $\pi_{\text{ref}}$ , prompt dataset  $\mathcal{D}$ , group size  $G$ , batch size  $B$ , search tool  $\mathcal{R}$ .
2: Initialize policy  $\pi_{\theta} \leftarrow \pi_{\theta_0}$ .
3: for each training step do
4:    $\pi_{\theta_{\text{old}}} \leftarrow \pi_{\theta}$ .
5:   Initialize training buffer  $\mathcal{B} \leftarrow \emptyset$ .
6:   while  $|\mathcal{B}| < B$  do
7:     Sample a prompt  $x \sim \mathcal{D}$ .
8:     Generate a group of  $G$  trajectories  $\{y_i\}_{i=1}^G$  using  $\pi_{\theta_{\text{old}}}$  and the search tool  $\mathcal{R}$ .
9:     Compute terminal rewards  $\{R_i\}_{i=1}^G = \{\text{ContainsAnswer}(y_i, y_{\text{gold}})\}_{i=1}^G$ .
10:    if  $0 < \sum_{i=1}^G R_i < G$  then ▷ Dynamic outcome-based filtering
11:      Add  $(x, \{y_i\}_{i=1}^G, \{R_i\}_{i=1}^G)$  to  $\mathcal{B}$ .
12:    end if
13:  end while
14:  for each  $(x, \{y_i\}, \{R_i\})$  in  $\mathcal{B}$  do
15:    Compute advantages  $\{\hat{A}_i\}_{i=1}^G$  via group normalization of  $\{R_i\}$ .
16:    Compute sequence-level importance ratios  $\{s_i(\theta)\}_{i=1}^G$  using Eq. 6.
17:    Compute the DSPO loss for the group using Eq. 11, applying masks to retrieved tokens.
18:  end for
19:  Update policy parameters  $\theta$  by taking a gradient step on the total loss from  $\mathcal{B}$ .
20: end for

```



Figure 2: **Validation subEM of DSPO across the seven QA benchmarks during training.** Held-out accuracy rises steadily on every benchmark and tracks the training reward, indicating that DSPO’s reward improvement reflects genuine generalization rather than reward hacking and that the learned search-and-reasoning policy holds across single-hop and multi-hop question-answering tasks.

Benchmarks and Baselines. To ensure a directly comparable evaluation, our experimental design follows the established protocol of Search-R1 (Jin et al., 2025b). We train on a composite dataset comprising the training splits of Natural Questions (NQ) (Kwiatkowski et al., 2019) and HotpotQA (Yang et al., 2018), and we evaluate on the held-out test sets of seven diverse QA benchmarks: NQ, TriviaQA (Joshi et al., 2017), PopQA (Mallen et al., 2022), HotpotQA, 2WikiMultiHopQA (Ho et al., 2020), Musique (Trivedi et al., 2022), and Bamboogle (Press et al., 2022). Because training draws only on NQ and HotpotQA, the remaining five benchmarks additionally serve as out-of-distribution tests of the learned policy. The suite spans single-hop and multi-hop reasoning, letting us characterize performance across a wide range of retrieval and reasoning difficulty.

Our comparison pairs external baselines with controlled internal ablations. The external baselines are the Qwen2.5-7B and 14B models trained with PPO and GRPO in the Search-R1 framework (Jin et al., 2025b;a). To isolate the contribution of each DSPO component, we define two ablations that each remove exactly one component: (1) **DSPO w/o**

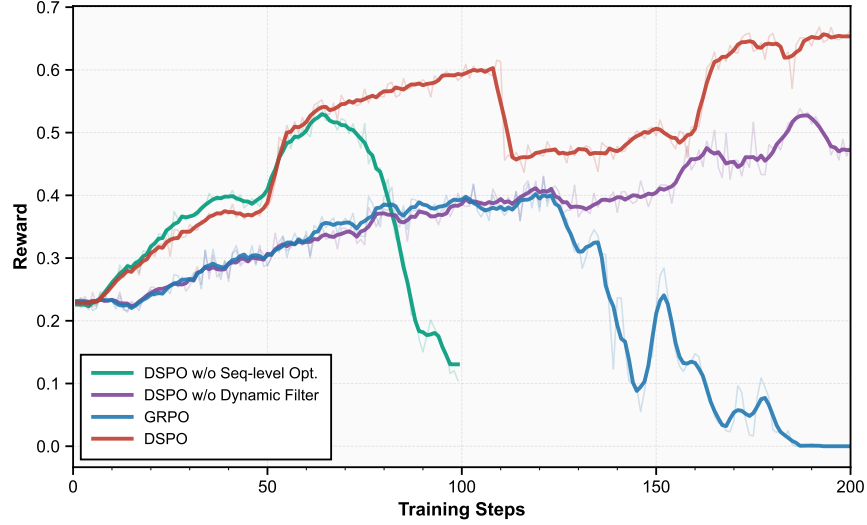


Figure 3: **Training reward dynamics of DSPO and its ablations.** The learning curves separate the algorithms sharply. DSPO (red) improves stably to the highest reward; the token-level variants (green, blue) undergo catastrophic policy collapse after a brief initial gain; and the sequence-level variant without the dynamic filter (purple) remains stable but plateaus at a markedly lower level. The contrast isolates the two components: sequence-level optimization supplies stability, while the dynamic filter supplies the signal density needed to reach a high reward.

Prompt Template. Answer the given question. You must conduct reasoning inside `<think>` and `</think>`. first every time you get new information. After reasoning, if you find you lack some knowledge, you can call a search engine by `<tool_call>` query `</tool_call>` and it will return the top searched results between `<tool_response>` and `</tool_response>`. You can search as many times as your want. If you find no further external knowledge needed, you can directly provide the answer inside `<answer>` and `</answer>`, without detailed illustrations. For example, `<answer>` Beijing `</answer>`. **Question:** ...

Table 1: The prompt template used in all our experiments, which instructs the model to interleave reasoning, tool-based search, and answer generation within the designated tags.

dynamic filter, which reduces to GSPO (Zheng et al., 2025), and (2) **DSPO w/o sequence-level opt.**, which reverts to the token-level policy DAPO (Yu et al., 2025). Because each ablation is itself a published algorithm, this design doubles as a direct comparison against GSPO and DAPO and lets us attribute performance differences unambiguously to the individual components.

All 7B training runs initialize from the Qwen2.5-7B-Instruct checkpoint. Our implementation builds on the VeRL framework (Sheng et al., 2025), adapting its Search-R1-like example code and scripts to our methodology. Because our setup uses a modified reward function, we do not rely on the originally published Search-R1 numbers for the primary comparison; instead, we retrain the PPO and GRPO baselines (Jin et al., 2025b;a) under identical conditions—the same training data, retriever, and reward—so that every reported difference reflects the optimization algorithm rather than a difference in environment or supervision. These retrained models constitute our primary external benchmarks, and for reference we also report the originally published Search-R1 EM scores in parentheses in Table 2.

Implementation and Evaluation. To separate the contribution of the optimization algorithm from that of retrieval quality, all RL experiments deliberately employ a single standard BM25 retriever shared across every method. Holding the retriever fixed and deliberately basic ensures that observed performance differences are attributable to the learned policy rather than to a stronger retrieval backbone; it also means our reported accuracies are a conservative lower bound on what the same algorithm could achieve with a stronger retriever. Across all methods, models are trained with a sparse binary reward defined by substring Exact Match (subEM) of the final answer, and subEM likewise serves as the primary evaluation metric, keeping the training and evaluation criteria consistent.

4.2 MAIN RESULTS AND ABLATION STUDY

Table 2 presents the main results and the ablation study together, juxtaposing DSPO against both the external Search-R1 baselines and its own ablated variants. Because the two components of DSPO act synergistically—each remedies a failure mode the other leaves open—reporting them in a single table lets the reader read DSPO’s overall standing and the role of each component from one side-by-side comparison.

Table 2: Comparison of DSPO against external baselines and ablation variants on seven QA benchmarks, reported as subEM accuracy. External baselines are the Search-R1 models (7B and 14B) trained with GRPO and PPO (Jin et al., 2025b;a); the ablations remove one component each (‘w/o dynamic filter’, equivalent to GSPO, and ‘w/o seq-level opt.’, equivalent to DAPO). Values in parentheses are the originally published Search-R1 EM scores; for a consistent comparison we retrained and re-evaluated every baseline under our adjusted reward, and all non-parenthetical values use that matched protocol. Best results per row are in **bold** and second-best are underlined. At 7B, DSPO attains the best or second-best score on every benchmark and the highest average overall, matching the average of the 14B GRPO baseline while surpassing it on four of the seven datasets.

Dataset	Search-R1				DSPO & Ablations (Ours, 7B)		
	GRPO (7B)	PPO (7B)	GRPO (14B)	PPO (14B)	w/o dyn. filter	w/o seq-lvl opt.	DSPO
NQ	0.423 (0.429)	(0.393)	<u>0.535</u> (0.482)	(0.424)	0.363	0.470	0.580
TriviaQA	0.658 (0.623)	(0.610)	0.760 (0.667)	(0.660)	0.515	0.695	<u>0.754</u>
PopQA	0.395 (0.427)	(0.397)	<u>0.477</u> (0.434)	(0.442)	0.277	0.430	0.498
HotpotQA	0.401 (0.386)	(0.370)	<u>0.563</u> (0.429)	(0.436)	0.330	0.438	0.613
2WikiMultiHopQA	0.357 (0.414)	(0.346)	0.611 (0.424)	(0.379)	0.285	0.398	<u>0.569</u>
Musique	0.122 (0.162)	(0.146)	<u>0.260</u> (0.191)	(0.210)	0.105	0.133	0.270
Bamboogle	0.280 (0.400)	(0.368)	0.504 (0.492)	(0.480)	0.288	0.280	<u>0.432</u>
Average	0.377 (0.396)	(0.385)	<u>0.530</u> (0.446)	(0.433)	0.313	0.406	0.531

Comparison with Baselines. The results in Table 2 establish DSPO’s advantage under matched conditions. Our DSPO-trained 7B agent attains an average subEM of 0.531—the highest of any method in the table—corresponding to a 34.1% relative improvement over the equally sized Search-R1 (GRPO, 7B) model at 0.377. Notably, this 7B agent edges out even the much larger Search-R1 14B models under both GRPO (0.530) and PPO (lower still), and it does so most emphatically on the hardest multi-hop benchmark, HotpotQA, where it reaches 0.613 against the GRPO-14B score of 0.563—a nearly 9% relative gain. The advantage is not uniform: on 2WikiMultiHopQA and Bamboogle the 14B GRPO model retains the lead, so DSPO’s 7B policy matches rather than dominates the 14B baseline in the aggregate. That a 7B policy nonetheless equals a 14B one on average, and surpasses it on the majority of benchmarks, indicates that the improvement flows from a more effective and stable learning algorithm rather than from raw model scale.

Analysis of Ablations. The ablation results in Table 2 show that both components are necessary and that neither suffices in isolation. Removing the dynamic filter (‘w/o dynamic filter’, i.e., GSPO) drops the average to 0.313: deprived of the filter, the sequence-level objective is starved of a useful gradient by the prevalence of homogeneous-reward batches, and it in fact trails the full 7B model on every one of the seven benchmarks. Ablating sequence-level optimization (‘w/o sequence-level opt.’, i.e., DAPO) yields 0.406; although this token-level variant improves on the filter-less one, it falls well short of the full DSPO model (0.531) and, as the next section shows, remains prone to catastrophic training instability. The two ablations thus fail for complementary reasons: sequence-level updates supply stability, the dynamic filter supplies signal density, and only their combination realizes the full performance. This complementary pattern is the central evidence for our claim that the contribution lies in the integration rather than in either component alone.

Beyond aggregate metrics, the trajectories DSPO produces exhibit recognizable search behaviors—recognition of irrelevant results, query reformulation, and multi-turn verification—which emerge from RL training alone, without any supervised demonstration of such strategies. We present representative examples in Appendix A; we characterize these as illustrative rather than as a systematic behavioral analysis, and a quantitative study of retrieval and reformulation quality remains future work.

4.3 ANALYSIS OF TRAINING DYNAMICS

To substantiate our claims of stability and efficiency, we analyze the training reward dynamics of DSPO, its ablations, and the key baselines. Figure 3 visualizes these trajectories, and the qualitative separation among them mirrors the final-accuracy gaps reported above.

DSPO (red) ascends smoothly, converging to the highest reward level and confirming the stability conferred by its sequence-level objective. In stark contrast, the **token-level methods**—DSPO w/o Seq-level Opt. (green) and vanilla GRPO (blue)—undergo catastrophic policy collapse early in training: after a brief initial gain their rewards plummet, a direct manifestation of the instability induced by high-variance token-level gradient updates. **DSPO w/o Dynamic Filter** (purple) retains sequence-level updates and so remains stable, but, lacking a dense learning signal, it saturates at a markedly lower ceiling. Read together, the curves separate the two failure modes cleanly: stability and signal density are each necessary and only jointly sufficient, and the synergy of sequence-level optimization and dynamic filtering is what yields policy optimization that is at once robust and high-performing.

To confirm that gains in training reward reflect generalization rather than reward hacking, we track validation subEM on the held-out benchmarks throughout training. As Figure 2 shows, DSPO’s validation accuracy on NQ, HotpotQA, and the remaining benchmarks rises in step with its reward curve. This close correspondence between training reward and held-out accuracy—including on the five benchmarks absent from training—indicates that the agent acquires a search-and-reasoning policy that generalizes rather than one that overfits the reward.

4.4 SCALABILITY AND GENERALIZATION ANALYSIS

To probe the robustness of our approach beyond the primary setting, we extend the evaluation along two axes: scaling the policy to a larger model, and transferring the algorithm to a different reasoning domain.

Scalability to Larger Models. We test whether DSPO’s stability benefits persist at larger parameter scales by training Qwen2.5-14B-Instruct. As Table 3 shows, the advantage not only persists but widens: the DSPO-trained 14B model attains an average subEM of 60.6%, exceeding the GRPO-14B baseline (53.0%) by a relative margin of **14.3%** and improving over the DSPO-trained 7B model on every benchmark. The gains hold across all seven datasets, ranging from +5.5% relative on TriviaQA to +38.8% on the demanding Musique benchmark. Two conclusions follow: DSPO converts additional capacity into commensurate performance rather than saturating, and its edge over the standard GRPO baseline is preserved—indeed amplified—as the model grows, which argues that the algorithm’s benefits are not an artifact of a particular model size.

Table 3: Scalability analysis on Qwen2.5-14B-Instruct, reported as subEM accuracy across the seven QA benchmarks, alongside the untrained Instruct model, the GRPO-14B baseline, and the DSPO-7B model for reference. The final column gives the relative gain of DSPO (14B) over the GRPO (14B) baseline. Best results per row are in **bold**; DSPO (14B) is best on every benchmark.

Dataset	Instruct (14B)	GRPO (14B)	DSPO (7B)	DSPO (14B)	Gain
NQ	0.345	0.535	0.580	0.629	+17.6%
HotpotQA	0.407	0.563	0.613	0.665	+18.1%
2WikiMQA	0.332	0.611	0.569	0.699	+14.4%
Bamboogle	0.328	0.504	0.432	0.544	+7.9%
PopQA	0.364	0.477	0.498	0.545	+14.3%
TriviaQA	0.643	0.760	0.754	0.802	+5.5%
Musique	0.151	0.260	0.270	0.361	+38.8%
Average	0.367	0.530	0.531	0.606	+14.3%

Generalization to Mathematical Reasoning. To assess whether DSPO’s benefits extend beyond agentic search, we apply it to single-turn mathematical reasoning—a setting with no retrieval tool, in which the trajectory is a single solution and the reward is verified answer correctness—using models from the Qwen2.5 and Qwen3 families. Table 4 reports the comparison against GRPO under matched training steps on Math500 and Olympiad-Bench. DSPO surpasses GRPO in both settings, by +2.6% relative on Math500 with Qwen2.5-Math-7B and by +2.7% on Olympiad-Bench with Qwen3-4B. The margins are modest and the study spans only two benchmarks and two models, so we

read this experiment as preliminary evidence rather than a definitive result; nonetheless, that the same optimization principles improve over GRPO once retrieval is removed altogether suggests that matching the optimization unit to the reward and guaranteeing signal density are not specific to search, and motivates a fuller investigation of DSPO in single-turn reasoning.

Table 4: Preliminary generalization to single-turn mathematical reasoning (no retrieval) on Math500 and Olympiad-Bench, comparing DSPO with GRPO at matched training steps. The final column reports the relative gain of DSPO over GRPO; best results per row are in **bold**.

Model	Benchmark	Steps	GRPO	DSPO	Gain
Qwen2.5-Math-7B	Math500	200	0.772	0.798	+2.6%
Qwen3-4B	Olympiad-Bench	100	0.728	0.755	+2.7%

5 CONCLUSION

In this work, we addressed the instability and sample-inefficiency that impede RL for autonomous LLM search agents. We introduced Dynamic-filter Sequence-level Policy Optimization (DSPO), which secures robust training by combining two complementary components—sequence-level optimization, which averts catastrophic policy collapse, and a dynamic outcome-based filter, which converts sparse rewards into a consistently informative learning signal—and we showed, through ablations that isolate each, that the two are separately necessary and jointly sufficient. Under a controlled protocol that holds the data, retriever, and reward fixed across methods, DSPO delivers the strongest performance across seven question-answering benchmarks—where a 7B model matches or exceeds 14B baselines—together with markedly superior training stability, and these benefits scale to a 14B policy.

By enabling robust training from environmental feedback alone, DSPO offers a practical blueprint for building capable LLM agents without costly expert demonstrations. Its principles are not confined to search: because sparse rewards and unstable policy gradients also arise elsewhere, our preliminary results on mathematical reasoning already show consistent gains over GRPO, and we expect related benefits in domains such as code generation. We are equally explicit about the study’s boundaries. Our evaluation rests on subEM over QA with a fixed BM25 retriever; the behavior of DSPO under dense or web-scale retrieval, on longer documents, and on non-QA tool-use tasks remains to be established, as does a systematic accounting of the extra rollout cost that dynamic filtering incurs and a broader, multi-seed characterization of variance. These are the natural next steps—integrating stronger retrievers, extending to multi-tool settings, and analyzing the filtering-induced sampling trade-off—that the stable foundation DSPO provides now makes tractable. We nonetheless regard its two core tenets—matching the optimization unit to the reward signal and guaranteeing signal density—as broadly useful ingredients for training the next generation of autonomous LLM agents.

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A APPENDIX

To illustrate the search strategies DSPO induces, Table A.1 presents representative trajectories generated by our model. These hand-selected examples are meant to make three learned competencies concrete rather than to quantify their frequency: (1) reformulating queries when an initial search returns irrelevant results, (2) verifying information through follow-up searches, and (3) extracting answers from retrieved evidence—each acquired through RL training alone, without supervised demonstration. A systematic, quantitative analysis of these behaviors across the benchmark suite is left to future work.

Table A.1: Representative trajectories illustrating the search and reasoning behaviors that DSPO learns through RL alone. The selected examples show (a) recognition of irrelevant results, (b) query reformulation, and (c) verification through follow-up search; they are illustrative and not a quantitative measure of how often each behavior occurs.

Example 1: Presidential Succession (<i>Query Refinement after Failure</i>)	
Question	Who is next in line if something happens to the president?
Gold	vice president
Turn 1	<code>search("who is next in line if something happens to the president")</code> → Retrieved: Documents about novels "Something Happened" and "Mosquitoes Buzz" Agent: "The queries were not directly about the line of succession... The relevant information I need is not present in the search results. Let's try searching again with a more specific query."
Turn 2	<code>search("line of succession for the president")</code> → Retrieved: Philippine and US presidential succession information Agent: "Based on the search results... the next person in line would be the Vice President."
Answer	✓ The Vice President
Example 2: University Identification (<i>Information Verification</i>)	
Question	Which American University is situated in Ithaca, New York?
Gold	Cornell University
Turn 1	<code>search(["American University in Ithaca", "University in Ithaca NY"])</code> → Retrieved: "Ithaca is home to Cornell University, an Ivy League school..." Agent: "Ithaca, New York is home to Cornell University, which is an Ivy League school. However... it is highly likely that the university in question is Cornell. Let me confirm this."
Turn 2	<code>search(["Cornell University location", "Cornell University New York"])</code> → Retrieved: Cornell history confirming establishment in Ithaca Agent: "Cornell University is indeed located in Ithaca, New York."
Answer	✓ Cornell University
Example 3: Biographical Information (<i>Direct Extraction</i>)	
Question	What is John Andrew Martin's occupation?
Gold	lawyer, attorney, politician
Turn 1	<code>search(["John Andrew Martin occupation", "John Andrew Martin job"])</code> → Retrieved: "American journalist, attorney, soldier, and politician..." Agent: "John Andrew Martin was an American journalist, attorney, soldier, and politician. He represented Colorado in the U.S. House of Representatives."
Answer	✓ Journalist, attorney, soldier, and politician